

Saturation Temperature & Pressure Lab

Aim

To study the relationship between the saturation temperature and pressure of water (i.e. temperature and pressure at which water vaporises) and investigate the accuracy of saturation data obtained using laboratory apparatus.

Theory

It is commonly understood that the temperature at which water undergoes a liquid-to-vapour phase change varies with pressure. For example, water boils at a lower temperature when at high altitudes, such as encountered on mountains. This relationship between pressure and temperature at which the liquid-to-vapour phase change occurs may be plotted on a graph. The resulting line is termed the saturation line. Saturation lines may be obtained for any liquid, although in this experiment water will be used. When plotted on a graph of temperature T against absolute pressure P_{abs} , the result is a smooth curve.

The Saturation Lab: Overview

Figure 1 shows the Armfield saturation pressure apparatus. The silver-coloured boiler can be seen in Figure 1, along with the orange power/control console which is used to adjust the amount of heat input. More details such as the heating element, valves, and sensor positions are shown in Figure 2.

The state of the water is checked through the viewing port in the boiler. The isolating valve is closed once the water is boiling and all air has escaped from the filling point, making it a closed system of constant volume with only a saturated water mixture of water inside. The heat input is increased as the water is kept boiling, and the experimental temperature, T_{exp} , and gauge pressure, P_1 , are read from the thermometer (T1) and pressure gauge (P) indicated in Figures 1 and 2.

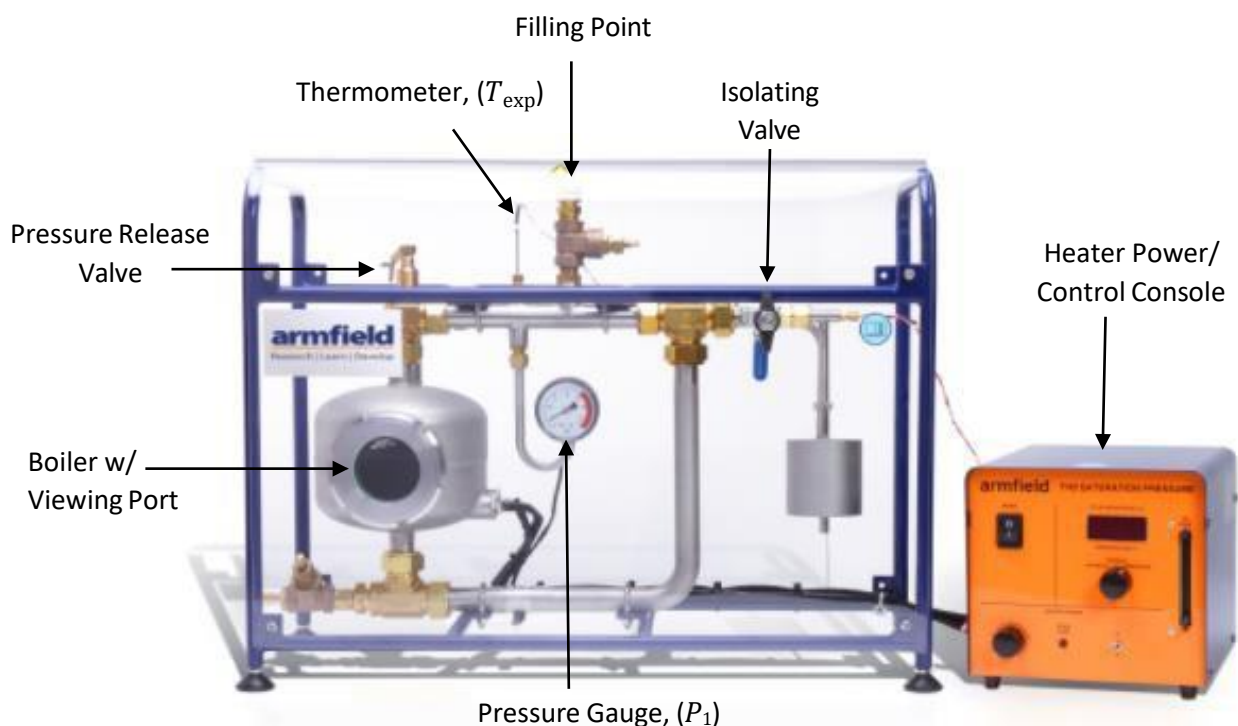


Figure 1: Armfield saturation pressure apparatus

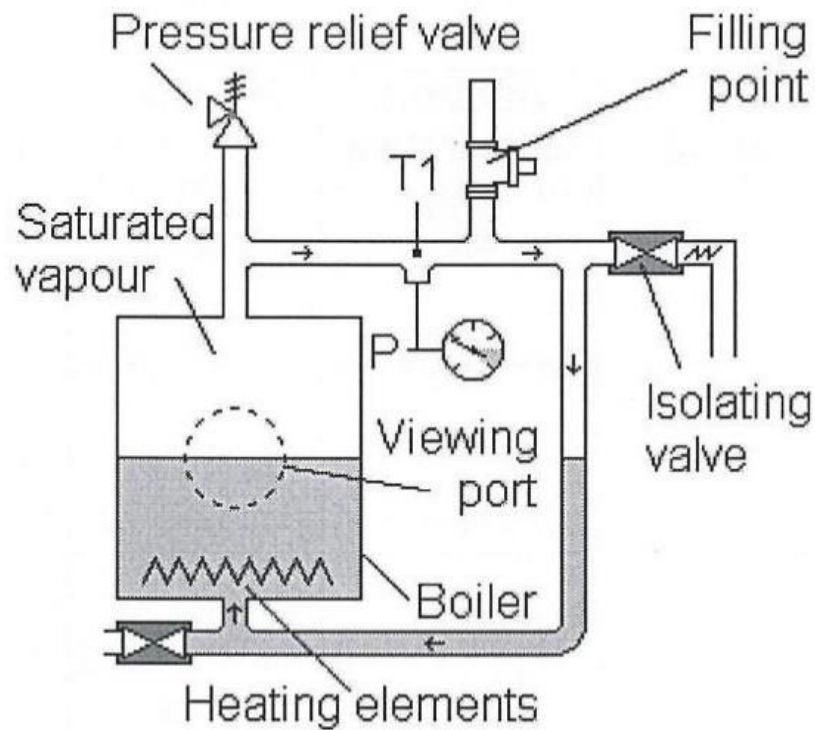


Figure 2: Labelled apparatus schematic diagram

Apparatus Set Up

1. Check that the isolating valve and the drain valve (at the base of the boiler) are both closed.
2. Check that the mains power to the orange console is switched off before filling the boiler.
3. Open the filling point valve at the top of the apparatus using the key provided. Fill the apparatus using purified or de-ionised water, until the water level is halfway up the sight-glass at the front of the boiler. Do not close the filling point valve until instructed later.
4. Switch on the mains power to the console and switch on the console itself.

Procedure

- Measure the atmospheric pressure using a barometer (e.g. use Fortin barometer in WZ111A).
- Allow the water to reach boiling point, indicated by intense movement at the surface and steam escaping from the filling point. Reduce the heater power slightly to maintain a steady but not excessive flow of steam. Wait until the temperature reading becomes steady, meaning that all air has been expelled.
- When heating fluid in the boiler, it takes time for heat to conduct through the apparatus to the pipework and the temperature probe. The platinum resistance thermometer (at T1) must reach the same temperature as the fluid before it will provide a valid reading. This time delay between the fluid reaching a given temperature and the sensor reaching the same value is known as thermal lag.
- Close the filling point valve then return the heater to maximum power.
- Allow the water to heat for two minutes, then lower the heater power. Record the temperature (T_{exp}) and gauge pressure (P_1) values when the temperature stabilizes. This allows heat to conduct from the fluid to the rest of the apparatus, and thus reduces the effect of thermal lag.
- Repeat the above step until gauge pressure reaches around 700 kPa (record 7 or 8 data points).
- Shutdown: Turn off the heating source. Slowly open the isolating valve and allow steam to bleed off. This will reduce both the pressure and temperature in the boiler. Once cooled, leave the isolating valve open.

Results

Atmospheric pressure on day of experiment: _____

TABLE 1: SATURATION LAB RESULTS

Gauge pressure, P_1 (kPa)	Absolute pressure, $P_{1,abs}$ (kPa)	Temperature, T_{exp} (°C)	Resistance, R (Ω)

Convert gauge pressures at each equilibrium stage to absolute pressures (using the atmospheric pressure measured on the day of the lab).

Lab Report Instructions

In this assessment you will write a lab report which includes the following:

- A **synopsis** of the entire report. This is a short summary of your entire report.
- Outline the **aim**, **equipment**, and experimental **method**. This may be adapted from the this lab instruction sheet.
- Analyse your **results** as follows:
 1. Convert the gauge pressures for each equilibrium stage to absolute pressures and plot a graph of *temperature (°C)* versus *absolute pressure (kPa)*. Use Excel to plot your graphs.
 2. On the same axes plot a graph of the saturation temperature/pressure values from the steam tables in the Appendix of your textbook ¹ (a copy is provided on Blackboard).
 3. The relationship between saturation temperature, (°C) and absolute saturation pressure, (kPa) can be considered to be of the form:

$$T_{sat} = AP_{sat}^n$$

where A and n are constants. Using the graphs you have plotted above, use the curve fitting capabilities of Excel (or similar) to determine the values of A and n for your experimental data AND for the steam table data. Clearly state these values in your Results.

4. To determine the errors between the experimental data and the steam table data it is necessary to determine the theoretical saturation temperature T_{th} for each experimental pressure reading. To do this, use the steam tables and **interpolate** to determine the theoretical saturation

temperature T_{th} for each recorded experimental pressure point $P_{1,abs}$. Record your results in a table similar to Table 3.

- For each data point, determine the error percentage between your experimental data, T_{exp} , and the interpolated table data, T_{th} , using the equation below.

$$Error \% = \frac{T_{exp} - T_{th}}{T_{th}} \times 100$$

From this determine an **average error** for your results.

TABLE 3: EXPERIMENTAL AND INTERPOLATED THEORETICAL RESULTS AND CORRESPONDING ERROR PERCENTAGE

Experimental Pressure $P_{1,abs}$ (kPa)	Experimental Temperature T_{exp} (°C)	Theoretical Temperature T_{th} (°C)	Error (%)

...

- Write a **discussion** section which includes the comments below:
 - Compare the experimental graphs of temperature vs pressure with the graph drawn using the steam table data.
 - Comment on the accuracy and usefulness of the equation of the graph from experimental data
 - Comment on the error percentage of your results - comment on both the overall trend in the error and also the average error percentage.
 - In the experiment, only the steam temperature (not the water temperature) was measured. Is it correct to assume that these will both be the same? How do the thermal lag and heat loss affect your experimental results?
 - What are the possible sources of error in your experiment?
- Finish the report with a **conclusion** which consists of a brief summary of the lab, and most important findings relative to the aim of the lab.

Submit your completed report by the due date via the link provided on Blackboard. See Blackboard for due date.

Notes on Report Formatting

Tables and Figures must be titled appropriately

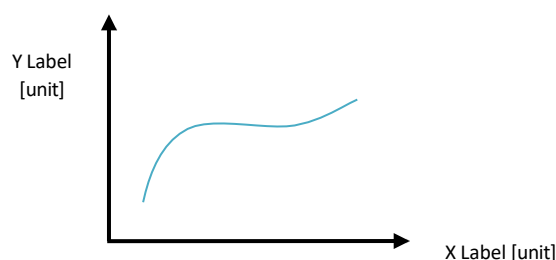


Figure 1: Example figure

Table 1: Example table

Heading (units)	Heading (units)
data	data
data	data
data	data
data	data
data	data
data	data

- Figure titles go underneath the figure (do not use the Excel default chart title at the top of your graph as your figure title).
- Table titles go on top of the table.
- If you do not refer to the figure/table in your text do not include it.
- Remember axis labels and units!*

No first-person pronouns

“We/I/She took eight temperature measurements” is inappropriate for a formal lab report.

“Eight temperature measurements were recorded” is more appropriate.

Paragraphs should not be excessively long

Your paragraphs are about one idea and developing that idea. If you move onto another statement/idea, then start a new paragraph.

Reference

¹ Cengel, Y., Boles, M., & Kanoglu, M. *Thermodynamics An Engineering Approach*, 9th edn in SI Units McGraw-Hill, 2020.